

```

# Potentiometric titration
# Packages 'gdata', 'econ' must be installed
# This software is totally free and comes to you without any warranty.
# You can use it freely and at your own risk.
rct(tstat(s))
library(gdata)
library(econ)
data=read.xls(file.choose(),sheet=1,headers=T)
nrow=(data)
x=data[,1]
y=data[,2]
# Titration curve
window(gettext("g,heightof:"))
windowsFont(AwWindowFont("Gentium Basic"))
par(mar=c(5,5,4,2)+0.1)
par(fontfamily="A")
plot(x,y,
      type="n",
      xlab=expression("Volume 0.1000 M AgNO3["* (mL)"),
      ylab="Electrode potential vs. SCE (V)",
      main="Potentiometric titration",
      col="blue",lty="n")
ntext("Raw data")
grid(x=0:100,y=0:100,lty="dotted",col="black",lwd=2)
points(x,y,type="n",pch=21,cex=1.25,bty="n",col="red")
# First derivative
window(gettext("g,heightof:"))
windowsFont(AwWindowFont("Gentium Basic"))
par(mar=c(5,5,4,2)+0.1)
par(fontfamily="A")
vector()
for(i in 2:n) {
  f[i]=((y[i]-y[i-1])/(x[i]-x[i-1]))
}
[100]--- Titration.R Top LI (ESS) [None] (ESC)
menu-bar ESS Font Lock ess-rl-keyword-fun-call

```

Figure 7: Emacs Speaks Statistics

according to <http://rmarkdown.rstudio.com>.

Emacs Speaks Statistics

With Emacs Speaks Statistics (<http://ess.r-project.org>), the user can edit R language commands in one GNU Emacs buffer and execute the code in a second (see Reference 9). Under Windows, an interesting way to do this is to use the GNU Emacs distribution prepared by Professor Vincent Goulet (Université Laval) and this is available for Mac OS too (<http://vgoulet.act.ulaval.ca/en/emacs>). I'm not an enthusiast of ESS because I think there's an interesting approach under GNU Linux but, when I'm using Windows, R has its own script editor and 'Edit > Run all' works fine for me.

There's another problem related to highlighting syntax. By default, the R language is poorly supported by Vim and not supported by GNU Emacs. The package ESS adds some syntax highlighting support to GNU Emacs for the R language. I'm not very satisfied with it because all the keywords have the same colour (Figure 7). This is probably due to the R keywords not being categorised. To have more effective syntax highlighting, I think it will be necessary to organise the keywords better, perhaps in a manner similar to Scilab (see OSFY, June 2015). Practically, it's not necessary to classify all the keywords (which could run into thousands) but only the keywords I'm using (200, more or less). For example, I can have different groups/colours for keywords about statistics (*lm*, *coef*, *predict* ...), plotting (*plot*, *points*, *abline* ...), ggplot (*ggplot*, *geom_something*, *ggtitle* ...), generic (*cbind*, *par*, *summary* ...) and more. This is more or less the same as in a 'human-spoken' language: it has a vocabulary (the keywords) with which it is possible to find various categories such as substantives, adjectives, verbs, adverbs, etc. Using this classification, I can build a syntax highlighting a file for Vim, GNU Emacs, jEdit, Notepad++, etc. An interesting explanation of

ESS is written by Rodney Sparapani as reported in Reference 10.

There's one more thing that must be said about the file encryption. Usually, my R files are not encrypted, but if a backup is done on a remote server, I prefer to encrypt them using GPG (GNU Privacy Guard, <http://www.gpg4win.org>). Yes, under Windows, because in my work experience I have encountered only one company that uses GNU Linux as a desktop system.

First, it's necessary to generate my key using GPA (GNU Privacy Assistant). Now Emacs can interact directly with GPG. In Emacs, visit a new file with a *.gpg* extension. For example, *my-script.R.gpg* and then place this line at the top of the file: *# -*- epa-file-encrypt-to: ("your@email.address") -*-*. When you try to save the file, Emacs will show a prompt in a buffer under it. Move the cursor to the line containing the key and hit 'm'. Move the cursor to 'OK' and hit '<return>'. Next time, when I open that *.gpg* file, I will be prompted for the password only once and then consecutive saves will be password-prompt-free. More details can be found on Emacs Stack Exchange (see Reference 11).

Without fear of contradiction, it can be said that R is today the *lingua franca* of statistics. In my opinion, it's one of the best examples of FOSS. The titration examples presented here are a bit unusual for an IT magazine, but do consider the fact that the automatic detection of one or more break-points in conductometric titrations is very useful in daily practice. I don't know if there are other free and open source software that do this as well as the *segmented* package for R.

R is also very well documented. There are a lot websites and books available to discover new things about it. **END** 

References

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- [10] <http://blog.revolutionanalytics.com/2014/03/emacs-ess-and-r-for-zombies.html>, last visited on 04/07/2015.
- [11] <http://emacs.stackexchange.com/questions/12212>, last visited on 04/07/2015.

By: Stefano Rizzi

The Author works in the areas of analytical chemistry and textile chemistry. He has been a Linux user since 1998.